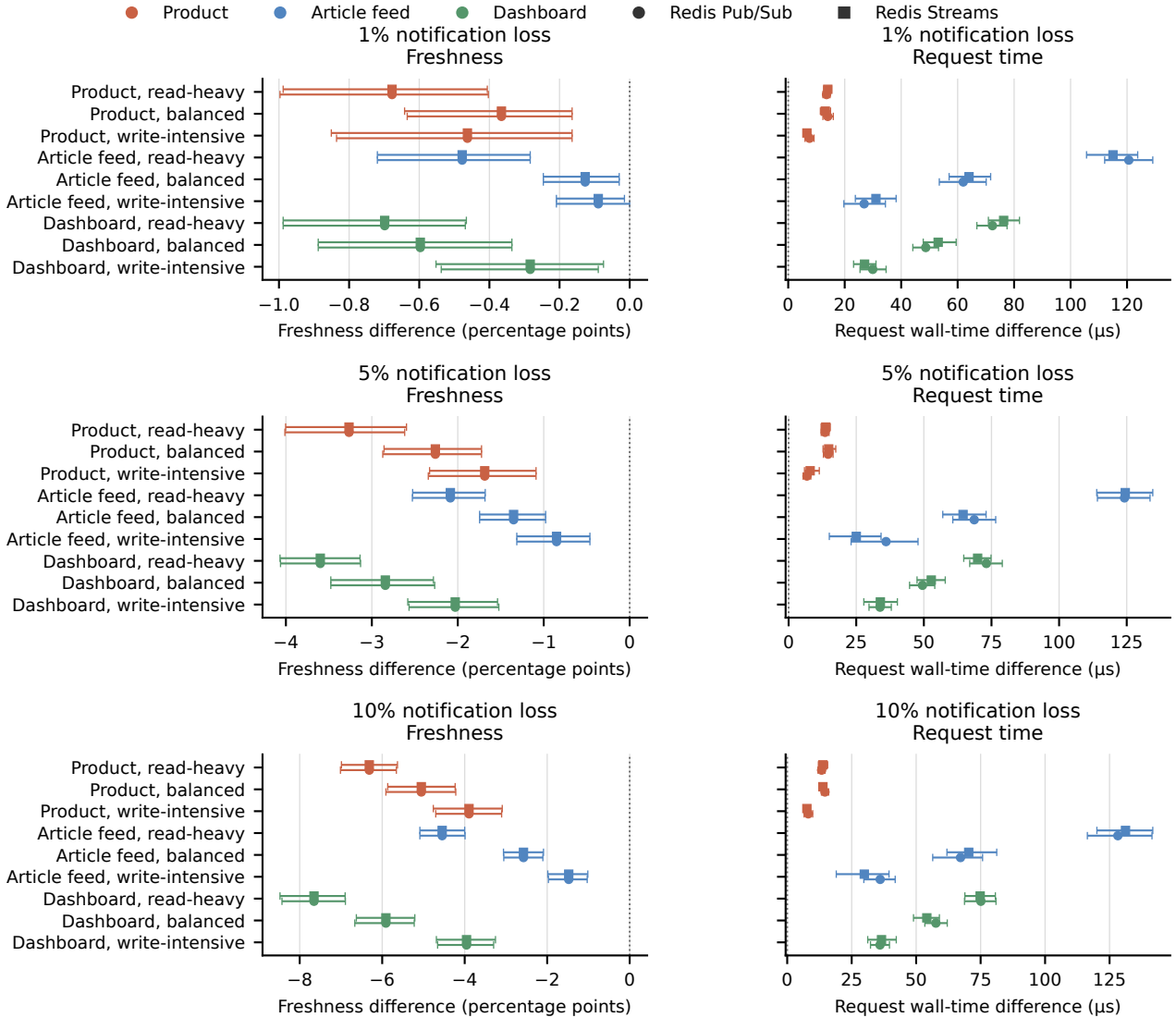
Bounded repair reduces freshness violations while increasing request time
Product
Article feed
Dashboard
Redis Pub/Sub
Redis Streams
1% notification loss
Freshness
Product, read-heavy
Product, balanced
Product, write-intensive
Article feed, read-heavy
Article feed, balanced
Article feed, write-intensive
Dashboard, read-heavy
Dashboard, balanced
Dashboard, write-intensive
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
Freshness difference (percentage points)
1% notification loss
Request time
0
20
40
60
80
100
120
Request wall-time difference (µs)
5% notification loss
Freshness
−4
−3
−2
−1
0
Freshness difference (percentage points)
5% notification loss
Request time
0
25
50
75
100
125
Request wall-time difference (µs)
10% notification loss
Freshness
−8
−6
−4
−2
0
Freshness difference (percentage points)
10% notification loss
Request time
0
25
50
75
100
125
Request wall-time difference (µs)